ROCSCIENCE

Case11
SLIDE - An Interactive Slope Stability Program
Rocscience Inc.
Date Created: 12/12/2016
Software Version: 9.041

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Slide2 Analysis Information

Case11

Project Summary

File Name:	Case11.slmd
Slide2 Modeler Version:	9.041
Compute Time:	00h:00m:01.652s
Project Title:	SLIDE - An Interactive Slope Stability Program
Author:	Rocscience Inc.
Company:	Rocscience Inc.
Date Created:	12/12/2016

General Settings

Units of Measurement:	Metric Units
Time Units:	days
Permeability Units:	meters/second
Data Output:	Standard
Failure Direction:	Right to Left

Analysis Options

Slices Type:	Vertical
Analysis Methods Used	
	Bishop simplified
	Spencer
Number of slices:	50
Tolerance:	0.0001
Maximum number of iterations:	75
Check malpha < 0.2:	Yes
Create Interslice boundaries at intersections with water tables and piezos:	Yes
Initial trial value of FS:	1
Steffensen Iteration:	Yes

Groundwater Analysis

Groundwater Method:	Water Surfaces
Pore Fluid Unit Weight [kN/m3]:	9.81
Advanced Groundwater Method:	None

Random Numbers

Pseudo-random Seed:	10116
Random Number Generation Method:	Park and Miller v.3

Surface Options

Surface Type:	Circular
Search Method:	Auto Refine Search
Divisions along slope:	20
Circles per division:	10
Number of iterations:	10
Divisions to use in next iteration:	50%
Composite Surfaces:	Disabled
Minimum Elevation:	Not Defined
Minimum Depth:	Not Defined
Minimum Area:	Not Defined
Minimum Weight:	Not Defined

Seismic Loading

Advanced seismic analysis:	No
Staged pseudostatic analysis:	No

Materials

soil 1	
Color	
Strength Type	Mohr-Coulomb
Unit Weight	19.5 kN/m3
Cohesion	0 kPa
Phi	38 °
Water Surface	None
Ru Value	0

soil 2	
Color	
Strength Type	Mohr-Coulomb
Unit Weight	19.5 kN/m3
Cohesion	5.3 kPa
Phi	23 °
Water Surface	None
Ru Value	0

soil 3	
Color	
Strength Type	Mohr-Coulomb
Unit Weight	19.5 kN/m3
Cohesion	7.2 kPa
Phi	20 °
Water Surface	None
Ru Value	0

Global Minimums

Method: bishop simplified

FS	1.403570
Center:	34.345, 43.177
Radius:	18.708
Left Slip Surface Endpoint:	29.921, 25.000
Right Slip Surface Endpoint:	51.171, 35.000
Resisting Moment:	14043.8 kN-m
Driving Moment:	10005.8 kN-m
Total Slice Area:	71.9005 m2
Surface Horizontal Width:	21.2495 m
Surface Average Height:	3.38363 m

Method: spencer

FS	1.373190
Center:	34.274, 43.129
Radius:	18.639
Left Slip Surface Endpoint:	29.945, 25.000
Right Slip Surface Endpoint:	51.047, 35.000
Resisting Moment:	13456.5 kN-m
Driving Moment:	9799.42 kN-m
Resisting Horizontal Force:	626.576 kN
Driving Horizontal Force:	456.293 kN
Total Slice Area:	70.4655 m2
Surface Horizontal Width:	21.1016 m
Surface Average Height:	3.33934 m

Global Minimum Support Data

No Supports Present

Valid and Invalid Surfaces

Method: bishop simplified

Number of Valid Surfaces:	10988
Number of Invalid Surfaces:	0

Method: spencer

Number of Valid Surfaces:	10816
Number of Invalid Surfaces:	172

Error Codes
Error Code -108 reported for 82 surfaces
Error Code -111 reported for 85 surfaces
Error Code -112 reported for 5 surfaces

Error Code Descriptions

The following errors were encountered during the computation:

-108 = Total driving moment or total driving force < 0.1. This is to limit the calculation of extremely high safety factors if the driving force is very small (0.1 is an arbitrary number).

-111 = Safety factor equation did not converge

-112 = The coefficient M-Alpha = cos(alpha)(1+tan(alpha)tan(phi)/F) < 0.2 for the final iteration of the safety factor calculation. This screens out some slip surfaces which may not be valid in the context of the analysis, in particular, deep seated slip surfaces with many high negative base angle slices in the passive zone.

Slice Data

Global Minimum Query (bishop simplified) - Safety Factor: 1.40357

Slice Number	Width [m]	Weight [kN]	Angle of Slice Base [deg]	Base Material	Base Cohesion [kPa]	Base Friction Angle [deg]	Shear Stress [kPa]	Shear Strength [kPa]	Base Normal Stress [kPa]	Pore Pressure [kPa]	Effective Normal Stress [kPa]	Base Vertical Stress [kPa]	Effective Vertical Stress [kPa]
1	0.435361	1.0464	-12.9932	soil 3	7.2	20	6.1192	8.58872	3.81548	0	3.81548	2.40352	2.40352
2	0.435361	3.67089	-11.6283	soil 3	7.2	20	7.72873	10.8478	10.0223	0	10.0223	8.43182	8.43182
3	0.435361	6.23404	-10.27	soil 3	7.2	20	9.27898	13.0237	16.0005	0	16.0005	14.3192	14.3192
4	0.435361	8.70686	-8.91763	soil 3	7.2	20	10.7535	15.0932	21.6865	0	21.6865	19.9992	19.9992
5	0.435361	11.0904	-7.57021	soil 3	7.2	20	12.1545	17.0598	27.0895	0	27.0895	25.4741	25.4741
6	0.435361	13.3857	-6.22699	soil 3	7.2	20	13.4843	18.9262	32.2175	0	32.2175	30.7462	30.7462
7	0.435361	15.5934	-4.8872	soil 3	7.2	20	14.7447	20.6952	37.0777	0	37.0777	35.817	35.817
8	0.435361	17.714	-3.55009	soil 3	7.2	20	15.9373	22.3691	41.6769	0	41.6769	40.6882	40.6882
9	0.435361	19.7482	-2.21491	soil 3	7.2	20	17.0636	23.95	46.0205	0	46.0205	45.3605	45.3605
10	0.435361	21.6961	-0.880936	soil 3	7.2	20	18.1251	25.4398	50.1134	0	50.1134	49.8347	49.8347
11	0.435361	23.5579	0.452563	soil 3	7.2	20	19.1226	26.8399	53.9602	0	53.9602	54.1112	54.1112
12	0.435361	25.3337	1.78631	soil 3	7.2	20	20.0573	28.1518	57.5644	0	57.5644	58.1899	58.1899
13	0.435361	27.0233	3.12102	soil 3	7.2	20	20.9299	29.3766	60.9296	0	60.9296	62.0708	62.0708
14	0.435361	28.6265	4.45743	soil 3	7.2	20	21.7413	30.5154	64.0585	0	64.0585	65.7533	65.7533
15	0.435361	30.1428	5.79628	soil 3	7.2	20	22.4919	31.569	66.9531	0	66.9531	69.2363	69.2363
16	0.435361	31.5718	7.13831	soil 3	7.2	20	23.1822	32.5379	69.6154	0	69.6154	72.5187	72.5187
17	0.435361	32.9127	8.4843	soil 3	7.2	20	23.8127	33.4228	72.0465	0	72.0465	75.5986	75.5986
18	0.435361	34.1647	9.83502	soil 3	7.2	20	24.3834	34.2238	74.2472	0	74.2472	78.4743	78.4743
19	0.435361	35.3267	11.1913	soil 3	7.2	20	24.8944	34.9411	76.2181	0	76.2181	81.1434	81.1434
20	0.435361	36.3976	12.5539	soil 3	7.2	20	25.3459	35.5748	77.9591	0	77.9591	83.6032	83.6032
21	0.435361	37.3759	13.9239	soil 3	7.2	20	25.7376	36.1245	79.4694	0	79.4694	85.8502	85.8502
22	0.435361	38.2601	15.3019	soil 3	7.2	20	26.0693	36.5901	80.7486	0	80.7486	87.8813	87.8813
23	0.435361	39.0485	16.6892	soil 3	7.2	20	26.3406	36.9709	81.7949	0	81.7949	89.692	89.692
24	0.435361	39.7389	18.0865	soil 3	7.2	20	26.5511	37.2663	82.6066	0	82.6066	91.2779	91.2779
25	0.435361	40.3291	19.4951	soil 3	7.2	20	26.7001	37.4755	83.1812	0	83.1812	92.6337	92.6337
26	0.435361	40.8166	20.9161	soil 3	7.2	20	26.7869	37.5973	83.5158	0	83.5158	93.7533	93.7533
27	0.435361	41.1985	22.3507	soil 3	7.2	20	26.8106	37.6305	83.607	0	83.607	94.6305	94.6305
28	0.435361	41.4716	23.8002	soil 3	7.2	20	26.77	37.5736	83.4508	0	83.4508	95.2579	95.2579
29	0.435361	41.6323	25.2661	soil 3	7.2	20	26.6641	37.4249	83.0423	0	83.0423	95.6271	95.6271
30	0.406148	38.8823	26.6996	soil 2	5.3	23	28.4077	39.8722	81.4469	0	81.4469	95.7342	95.7342
31	0.406148	38.823	28.1008	soil 2	5.3	23	28.1401	39.4966	80.5622	0	80.5622	95.5882	95.5882
32	0.406148	38.6618	29.5207	soil 2	5.3	23	27.8031	39.0236	79.4478	0	79.4478	95.1913	95.1913
33	0.406148	38.3945	30.9607	soil 2	5.3	23	27.3949	38.4507	78.0983	0	78.0983	94.5333	94.5333
34	0.406148	38.0164	32.4229	soil 2	5.3	23	26.9138	37.7754	76.5072	0	76.5072	93.6023	93.6023
35	0.406148	37.522	33.9091	soil 2	5.3	23	26.3574	36.9945	74.6675	0	74.6675	92.385	92.385
36	0.406148	36.9053	35.4218	soil 2	5.3	23	25.7234	36.1046	72.5711	0	72.5711	90.8665	90.8665
37	0.412928	36.7555	36.9766	soil 1	0	38	34.9148	49.0054	62.7241	0	62.7241	89.012	89.012
38	0.412928	35.8402	38.577	soil 1	0	38	33.4585	46.9614	60.1079	0	60.1079	86.7955	86.7955
39	0.412928	34.7711	40.214	soil 1	0	38	31.8725	44.7353	57.2585	0	57.2585	84.2062	84.2062
40	0.412928	33.5368	41.8915	soil 1	0	38	30.1533	42.3223	54.1701	0	54.1701	81.2171	81.2171
41	0.412928	32.1241	43.6144	soil 1	0	38	28.2971	39.7169	50.8353	0	50.8353	77.7958	77.7958
42	0.412928	30.5175	45.3882	soil 1	0	38	26.2995	36.9132	47.2467	0	47.2467	73.905	73.905
43	0.412928	28.6982	47.2197	soil 1	0	38	24.1558	33.9043	43.3956	0	43.3956	69.4994	69.4994
44	0.412928	26.6438	49.1169	soil 1	0	38	21.8607	30.683	39.2724	0	39.2724	64.5241	64.5241
45	0.412928	24.3264	51.0898	soil 1	0	38	19.4086	27.2413	34.8672	0	34.8672	58.9118	58.9118
46	0.412928	21.7109	53.1511	soil 1	0	38	16.7936	23.571	30.1695	0	30.1695	52.5781	52.5781
47	0.412928	18.7527	55.3168	soil 1	0	38	14.0098	19.6638	25.1686	0	25.1686	45.414	45.414
48	0.412928	14.8123	57.6086	soil 1	0	38	10.6357	14.9279	19.1068	0	19.1068	35.8715	35.8715
49	0.412928	9.32841	60.056	soil 1	0	38	6.39524	8.97617	11.489	0	11.489	22.5909	22.5909
50	0.412928	3.22121	62.7018	soil 1	0	38	2.0891	2.9322	3.75304	0	3.75304	7.80091	7.80091

Global Minimum Query (spencer) - Safety Factor: 1.37319

Slice Number	Width [m]	Weight [kN]	Angle of Slice Base [deg]	Base Material	Base Cohesion [kPa]	Base Friction Angle [deg]	Shear Stress [kPa]	Shear Strength [kPa]	Base Normal Stress [kPa]	Pore Pressure [kPa]	Effective Normal Stress [kPa]	Base Vertical Stress [kPa]	Effective Vertical Stress [kPa]
1	0.431594	1.10195	-12.75	soil 3	7.2	20	7.28665	10.006	7.70931	0	7.70931	6.0605	6.0605
2	0.431594	3.68025	-11.3932	soil 3	7.2	20	9.26752	12.7261	15.1827	0	15.1827	13.3152	13.3152
3	0.431594	6.18403	-10.0427	soil 3	7.2	20	11.1102	15.2565	22.135	0	22.135	20.1674	20.1674
4	0.431594	8.59968	-8.69795	soil 3	7.2	20	12.8138	17.5958	28.5622	0	28.5622	26.6019	26.6019
5	0.431594	10.9282	-7.35796	soil 3	7.2	20	14.3878	19.7572	34.5006	0	34.5006	32.6427	32.6427
6	0.431594	13.1705	-6.02201	soil 3	7.2	20	15.8408	21.7524	39.9824	0	39.9824	38.3113	38.3113
7	0.431594	15.3272	-4.68934	soil 3	7.2	20	17.18	23.5914	45.0352	0	45.0352	43.626	43.626
8	0.431594	17.399	-3.35921	soil 3	7.2	20	18.4122	25.2835	49.6839	0	49.6839	48.6032	48.6032
9	0.431594	19.3861	-2.03089	soil 3	7.2	20	19.5431	26.8364	53.9504	0	53.9504	53.2574	53.2574
10	0.431594	21.289	-0.703666	soil 3	7.2	20	20.5778	28.2572	57.8542	0	57.8542	57.6014	57.6014
11	0.431594	23.1077	0.623184	soil 3	7.2	20	21.5208	29.5522	61.4122	0	61.4122	61.6463	61.6463
12	0.431594	24.8423	1.95037	soil 3	7.2	20	22.3764	30.727	64.64	0	64.64	65.402	65.402
13	0.431594	26.4926	3.2786	soil 3	7.2	20	23.148	31.7866	67.551	0	67.551	68.877	68.877
14	0.431594	28.0583	4.6086	soil 3	7.2	20	23.8389	32.7353	70.1576	0	70.1576	72.0792	72.0792
15	0.431594	29.5391	5.94109	soil 3	7.2	20	24.452	33.5772	72.4706	0	72.4706	75.0152	75.0152
16	0.431594	30.9343	7.27682	soil 3	7.2	20	24.9898	34.3157	74.4997	0	74.4997	77.6907	77.6907
17	0.431594	32.2434	8.61654	soil 3	7.2	20	25.4547	34.9541	76.2536	0	76.2536	80.1108	80.1108
18	0.431594	33.4654	9.96103	soil 3	7.2	20	25.8486	35.495	77.7399	0	77.7399	82.2796	82.2796
19	0.431594	34.5993	11.3111	soil 3	7.2	20	26.1734	35.9411	78.9655	0	78.9655	84.2007	84.2007
20	0.431594	35.644	12.6675	soil 3	7.2	20	26.4307	36.2944	79.9362	0	79.9362	85.8769	85.8769
21	0.431594	36.5981	14.0313	soil 3	7.2	20	26.6219	36.5569	80.6573	0	80.6573	87.3103	87.3103
22	0.431594	37.46	15.4031	soil 3	7.2	20	26.7482	36.7303	81.1338	0	81.1338	88.5031	88.5031
23	0.431594	38.228	16.7841	soil 3	7.2	20	26.8105	36.8159	81.3692	0	81.3692	89.4557	89.4557
24	0.431594	38.9001	18.1752	soil 3	7.2	20	26.8099	36.8151	81.3669	0	81.3669	90.1687	90.1687
25	0.431594	39.4741	19.5775	soil 3	7.2	20	26.7471	36.7289	81.13	0	81.13	90.6424	90.6424
26	0.431594	39.9475	20.9922	soil 3	7.2	20	26.6227	36.558	80.6603	0	80.6603	90.8757	90.8757
27	0.431594	40.3175	22.4203	soil 3	7.2	20	26.4371	36.3031	79.9601	0	79.9601	90.8676	90.8676
28	0.431594	40.5809	23.8634	soil 3	7.2	20	26.1905	35.9646	79.03	0	79.03	90.616	90.616
29	0.431594	40.7343	25.3227	soil 3	7.2	20	25.8834	35.5428	77.8711	0	77.8711	90.1186	90.1186
30	0.394371	37.26	26.7353	soil 2	5.3	23	27.466	37.716	76.3675	0	76.3675	90.2026	90.2026
31	0.394371	37.2028	28.1012	soil 2	5.3	23	27.0034	37.0808	74.871	0	74.871	89.2902	89.2902
32	0.394371	37.0521	29.4847	soil 2	5.3	23	26.4828	36.3659	73.1866	0	73.1866	88.1605	88.1605
33	0.394371	36.804	30.8874	soil 2	5.3	23	25.904	35.5711	71.3141	0	71.3141	86.8096	86.8096
34	0.394371	36.4543	32.311	soil 2	5.3	23	25.2668	34.6961	69.2527	0	69.2527	85.2325	85.2325
35	0.394371	35.9981	33.7573	soil 2	5.3	23	24.5708	33.7404	67.0013	0	67.0013	83.4235	83.4235
36	0.394371	35.4302	35.2285	soil 2	5.3	23	23.8156	32.7034	64.5585	0	64.5585	81.3764	81.3764
37	0.416058	36.6314	36.7689	soil 1	0	38	33.5193	46.0283	58.9136	0	58.9136	83.9608	83.9608
38	0.416058	35.7211	38.383	soil 1	0	38	31.7729	43.6302	55.8441	0	55.8441	81.0116	81.0116
39	0.416058	34.6541	40.0341	soil 1	0	38	29.9288	41.0979	52.6028	0	52.6028	77.7464	77.7464
40	0.416058	33.4188	41.7262	soil 1	0	38	27.9872	38.4318	49.1904	0	49.1904	74.1491	74.1491
41	0.416058	32.0018	43.4642	soil 1	0	38	25.9484	35.6321	45.607	0	45.607	70.2003	70.2003
42	0.416058	30.3872	45.2537	soil 1	0	38	23.8128	32.6995	41.8534	0	41.8534	65.878	65.878
43	0.416058	28.5558	47.1017	soil 1	0	38	21.5808	29.6345	37.9303	0	37.9303	61.1553	61.1553
44	0.416058	26.4845	49.0163	soil 1	0	38	19.2531	26.4381	33.8391	0	33.8391	55.9999	55.9999
45	0.416058	24.1448	51.0077	soil 1	0	38	16.831	23.1121	29.5822	0	29.5822	50.3724	50.3724
46	0.416058	21.5008	53.089	soil 1	0	38	14.3167	19.6595	25.163	0	25.163	44.2234	44.2234
47	0.416058	18.5063	55.2766	soil 1	0	38	11.714	16.0856	20.5886	0	20.5886	37.4911	37.4911
48	0.416058	14.8759	57.5926	soil 1	0	38	8.89513	12.2147	15.6341	0	15.6341	29.6466	29.6466
49	0.416058	9.48411	60.0676	soil 1	0	38	5.31294	7.29567	9.33804	0	9.33804	18.5654	18.5654
50	0.416058	3.27642	62.7459	soil 1	0	38	1.70004	2.33448	2.988	0	2.988	6.28825	6.28825

Interslice Data

Global Minimum Query (bishop simplified) - Safety Factor: 1.40357

Slice Number	X coordinate [m]	Y coordinate - Bottom [m]	Interslice Normal Force [kN]	Interslice Shear Force [kN]	Interslice Force Angle [deg]
1	29.9213	25	0	0	0
2	30.3566	24.8995	3.04735	0	0
3	30.792	24.81	7.31005	0	0
4	31.2273	24.7311	12.6119	0	0
5	31.6627	24.6628	18.775	0	0
6	32.0981	24.6049	25.634	0	0
7	32.5334	24.5574	33.035	0	0
8	32.9688	24.5202	40.8345	0	0
9	33.4041	24.4932	48.8987	0	0
10	33.8395	24.4763	57.1025	0	0
11	34.2749	24.4696	65.3289	0	0
12	34.7102	24.4731	73.4686	0	0
13	35.1456	24.4866	81.4191	0	0
14	35.581	24.5104	89.0848	0	0
15	36.0163	24.5443	96.3761	0	0
16	36.4517	24.5885	103.209	0	0
17	36.887	24.643	109.506	0	0
18	37.3224	24.708	115.195	0	0
19	37.7578	24.7835	120.206	0	0
20	38.1931	24.8696	124.479	0	0
21	38.6285	24.9665	127.956	0	0
22	39.0638	25.0745	130.584	0	0
23	39.4992	25.1936	132.315	0	0
24	39.9346	25.3241	133.106	0	0
25	40.3699	25.4663	132.92	0	0
26	40.8053	25.6204	131.724	0	0
27	41.2407	25.7868	129.49	0	0
28	41.676	25.9658	126.196	0	0
29	42.1114	26.1578	121.826	0	0
30	42.5467	26.3633	116.371	0	0
31	42.9529	26.5676	111.272	0	0
32	43.359	26.7845	105.23	0	0
33	43.7652	27.0144	98.2503	0	0
34	44.1713	27.2581	90.3473	0	0
35	44.5775	27.5161	81.5412	0	0
36	44.9836	27.7891	71.861	0	0
37	45.3898	28.078	61.3451	0	0
38	45.8027	28.3889	56.2615	0	0
39	46.2156	28.7182	50.2801	0	0
40	46.6286	29.0673	43.4508	0	0
41	47.0415	29.4377	35.838	0	0
42	47.4544	29.8312	27.5228	0	0
43	47.8673	30.2497	18.607	0	0
44	48.2803	30.6959	9.2172	0	0
45	48.6932	31.1729	-0.488118	0	0
46	49.1061	31.6845	-10.3105	0	0
47	49.519	32.2355	-19.9991	0	0
48	49.932	32.8322	-29.2325	0	0
49	50.3449	33.4831	-37.2771	0	0
50	50.7578	34.1999	-42.8719	0	0
51	51.1708	35	0	0	0

Global Minimum Query (spencer) - Safety Factor: 1.37319

Slice Number	X coordinate [m]	Y coordinate - Bottom [m]	Interslice Normal Force [kN]	Interslice Shear Force [kN]	Interslice Force Angle [deg]
1	29.9449	25	0	0	0
2	30.3765	24.9023	3.89772	1.51506	21.2413
3	30.8081	24.8154	9.21792	3.58304	21.2413
4	31.2397	24.7389	15.7048	6.10453	21.2413
5	31.6713	24.6729	23.121	8.98723	21.2413
6	32.1029	24.6172	31.2534	12.1483	21.2413
7	32.5345	24.5716	39.9104	15.5134	21.2414
8	32.9661	24.5362	48.9195	19.0152	21.2413
9	33.3976	24.5109	58.1246	22.5933	21.2413
10	33.8292	24.4956	67.3848	26.1928	21.2413
11	34.2608	24.4903	76.5726	29.7641	21.2413
12	34.6924	24.495	85.5724	33.2624	21.2413
13	35.124	24.5097	94.2798	36.647	21.2413
14	35.5556	24.5344	102.6	39.8811	21.2413
15	35.9872	24.5692	110.448	42.9315	21.2413
16	36.4188	24.6141	117.746	45.7684	21.2413
17	36.8504	24.6692	124.425	48.3647	21.2414
18	37.282	24.7346	130.424	50.6965	21.2414
19	37.7136	24.8104	135.688	52.7424	21.2413
20	38.1452	24.8968	140.167	54.4835	21.2413
21	38.5768	24.9938	143.82	55.9033	21.2413
22	39.0084	25.1016	146.61	56.9879	21.2413
23	39.44	25.2205	148.507	57.7252	21.2413
24	39.8716	25.3507	149.485	58.1057	21.2414
25	40.3031	25.4924	149.527	58.1218	21.2413
26	40.7347	25.6459	148.618	57.7684	21.2413
27	41.1663	25.8115	146.75	57.0424	21.2413
28	41.5979	25.9896	143.922	55.9429	21.2412
29	42.0295	26.1805	140.136	54.4715	21.2413
30	42.4611	26.3847	135.404	52.6321	21.2413
31	42.8555	26.5834	131.065	50.9456	21.2413
32	43.2499	26.794	125.948	48.9564	21.2413
33	43.6442	27.0169	120.072	46.6725	21.2413
34	44.0386	27.2529	113.464	44.1039	21.2413
35	44.433	27.5023	106.155	41.2631	21.2414
36	44.8273	27.7659	98.1849	38.1649	21.2413
37	45.2217	28.0443	89.5979	34.8271	21.2413
38	45.6378	28.3552	85.2275	33.1283	21.2413
39	46.0538	28.6848	80.0425	31.1129	21.2413
40	46.4699	29.0343	74.1078	28.806	21.2413
41	46.8859	29.4054	67.5006	26.2378	21.2413
42	47.302	29.7997	60.3122	23.4436	21.2413
43	47.7181	30.2195	52.6512	20.4658	21.2414
44	48.1341	30.6672	44.6463	17.3542	21.2413
45	48.5502	31.1461	36.4512	14.1687	21.2413
46	48.9662	31.66	28.2505	10.9811	21.2413
47	49.3823	32.214	20.2688	7.87858	21.2413
48	49.7984	32.8143	12.7823	4.96855	21.2414
49	50.2144	33.4697	6.23633	2.42409	21.2413
50	50.6305	34.1923	1.69912	0.660453	21.2413
51	51.0465	35	0	0	0

Discharge Sections

Entity Information

Group 1

Shared Entities

Type	Coordinates (x,y)
External Boundary	20, 20 70, 20 70, 24 70, 31 70, 35 50, 35 30, 25 20, 25
Material Boundary	40, 27 52, 24 70, 24
Material Boundary	30, 25 40, 27 50, 29 54, 31 70, 31